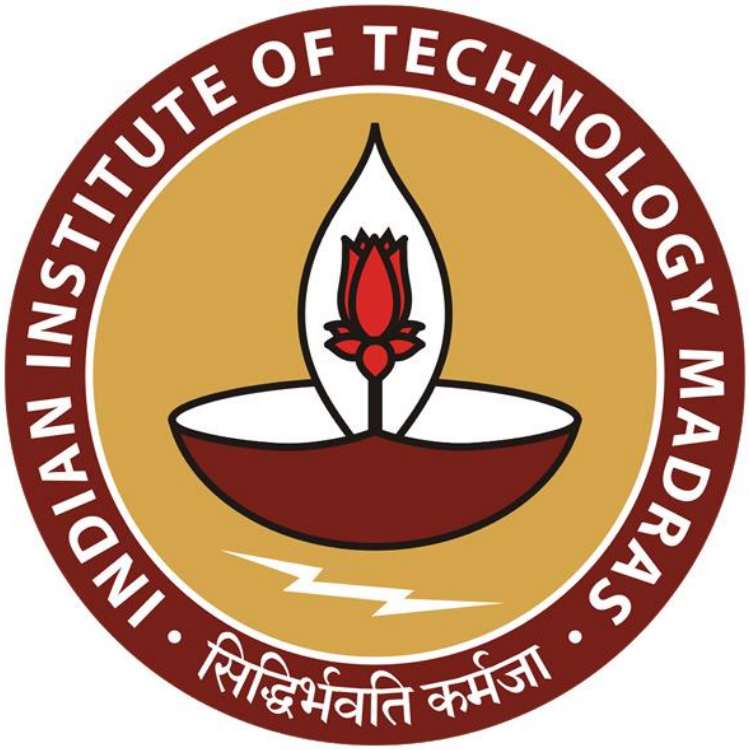
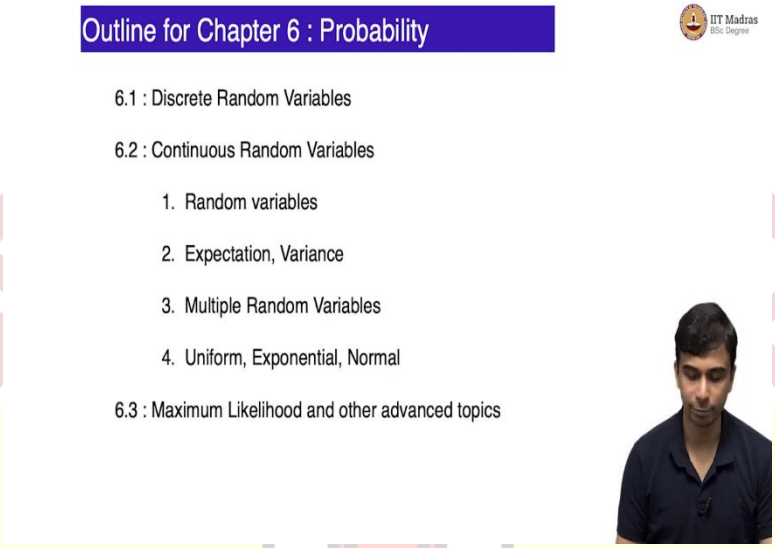


IIT Madras

ONLINE DEGREE

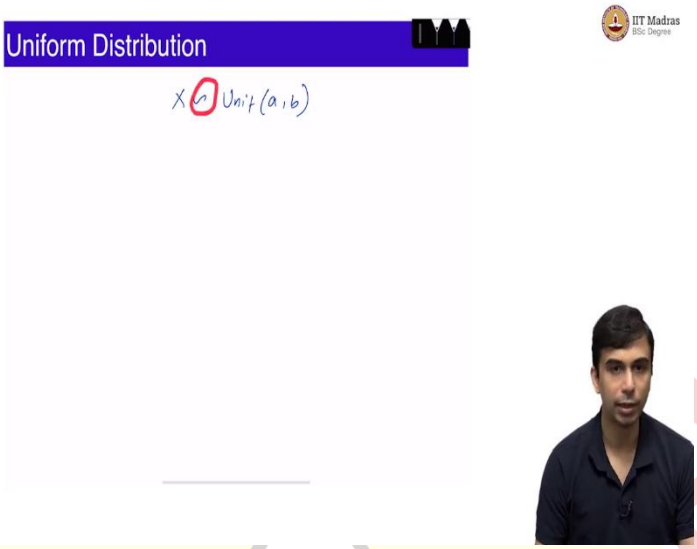
Professor Harish Guruprasad Ramaswamy
Department of Computer Sciences and Engineering
Indian Institute of Technology, Madras
Uniform, Exponential, Normal

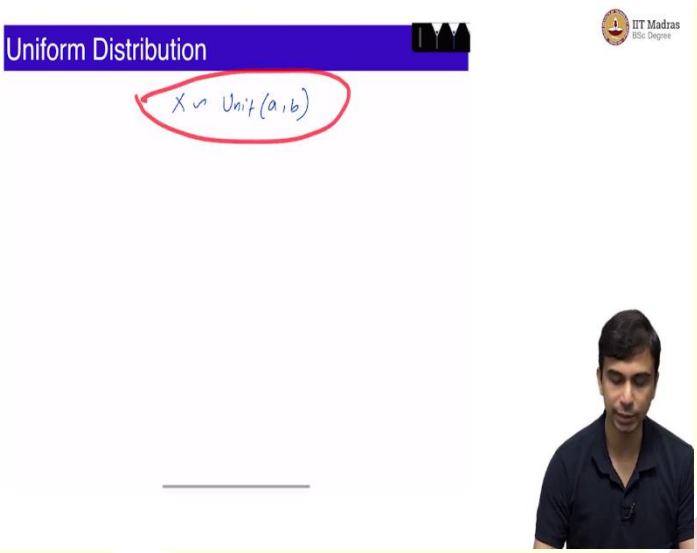
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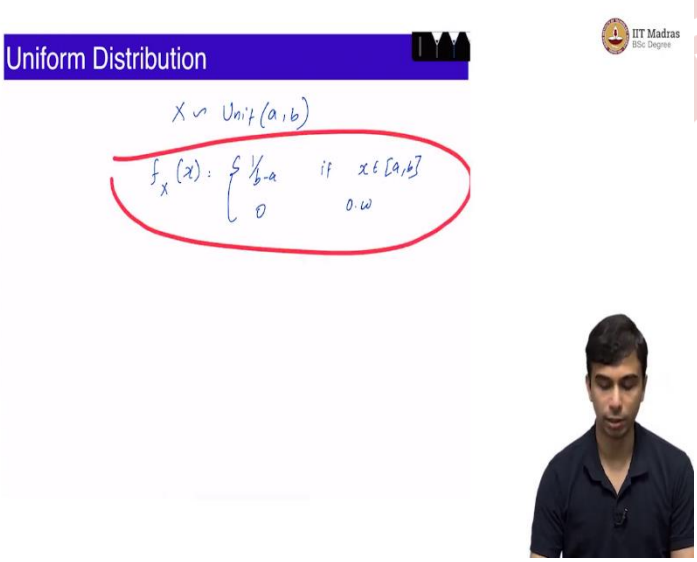


Hello, everyone, and welcome to another lecture on probability. In this lecture, we will continue with the continuous random variables. And, in particular, we will take a look at some standard random variable examples and study their properties. In particular, we will take a look at the uniform distribution, the exponential distribution and the normal distribution. These are very common continuous random variable of these distributions, which we will give the probability density function of these random many of these distributions. And we will see some nice properties that these distributions have.

(Refer Slide Time: 00:52)







Uniform Distribution

$X \sim Unif(a, b)$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in [a,b] \\ 0 & \text{o.w} \end{cases}$$



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$$E[X] = \frac{a+b}{2}$$

$$Var[X] = \frac{(b-a)^2}{12}$$



Uniform Distribution

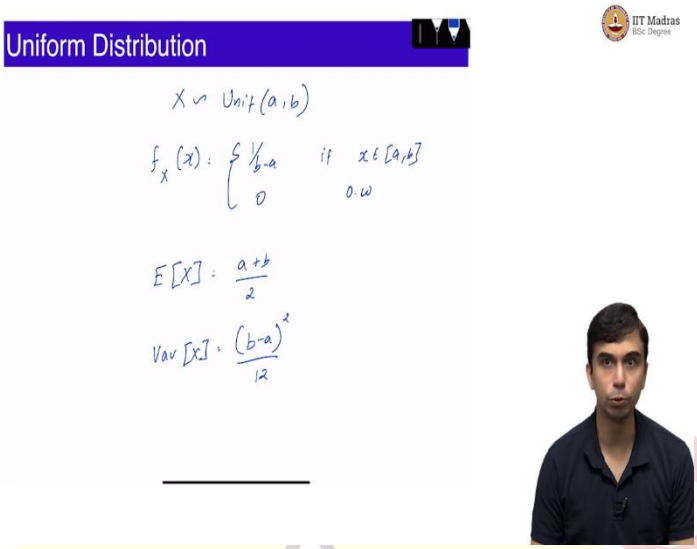
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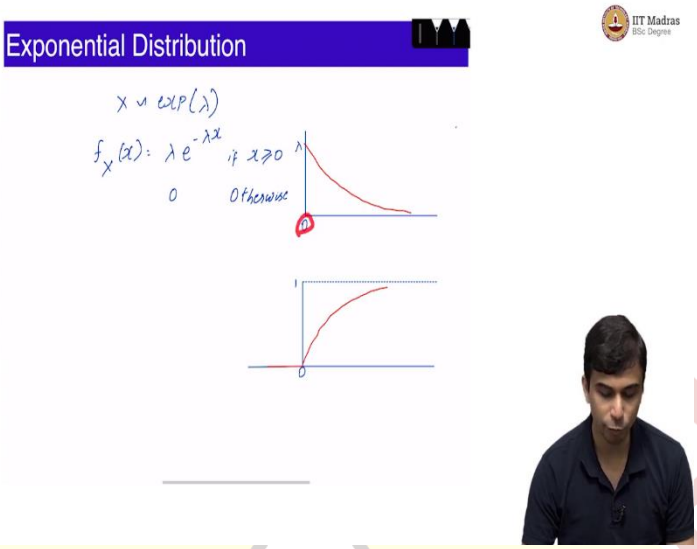


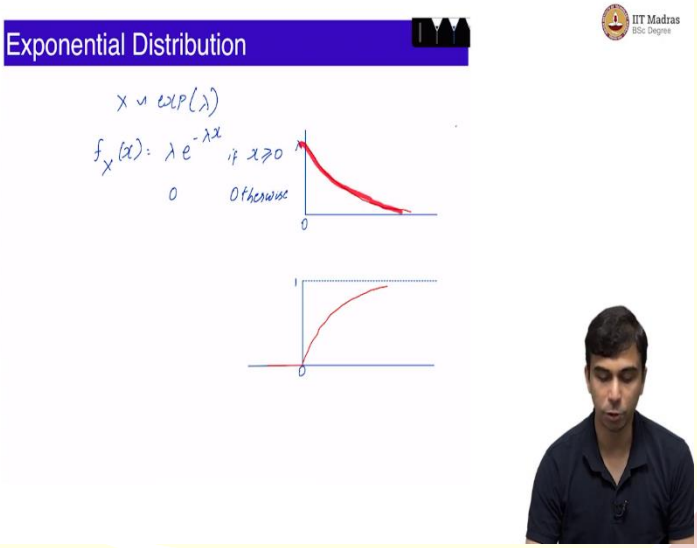
The first is the is an example you have seen already several times, which is a uniform distribution. We will often write X is distributed as uniform of a, b, you might see it written this way, where this wave symbol indicates distributed as this particular statement is to be read as x is the random variable X is distributed as uniform a , b. So, what does that mean?

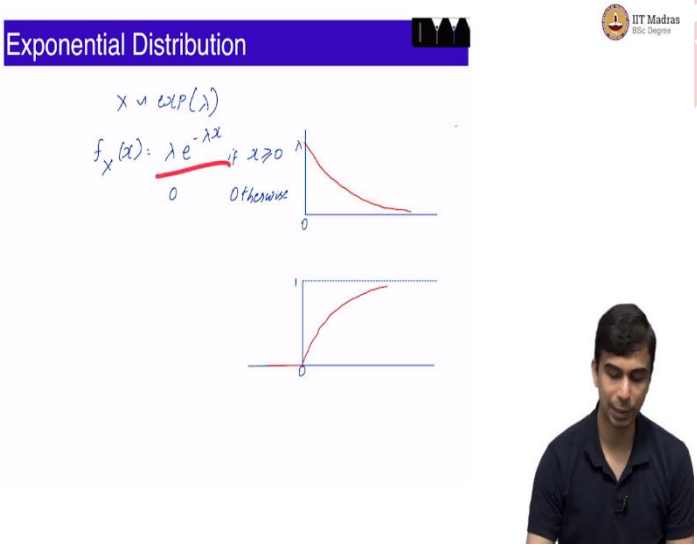
Well, that means the density function, $f_X(x)$ is simply equal to 1/(b – a), if x belongs to a to b, and 0, otherwise. This is the density function associated with this distribution. We have already seen this several times, but I am just making it giving you a notation of uniform a , b. And we also know what the expectation and variance of this random variable are, we just write that down again for completeness, the expected value of this random variable is simply (a + b)/2.

And the variance of this random variable is equal to (b – a) the whole square by 12. We derived both these things in a when we introduced expectation and variance. But it is very easy to derive these from first principles again, if you want. So that is all we have for the uniform distribution.

(Refer Slide Time: 02:30)







Exponential Distribution

$X \sim exp(\lambda)$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



$$\int_0^\infty \lambda e^{-\lambda x} dx = \lambda \left[\frac{-1}{\lambda} e^{-\lambda x} \right]_0^\infty$$

$$= \lambda \cdot \frac{1}{\lambda} = 1$$

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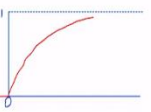
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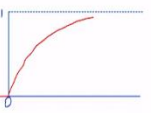
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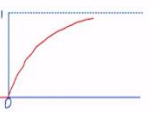
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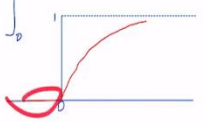
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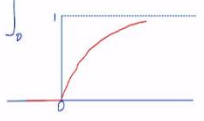
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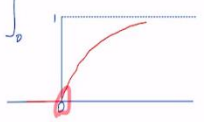
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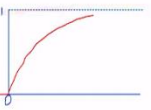
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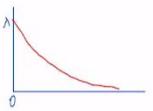
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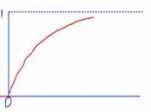
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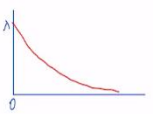
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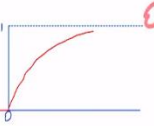
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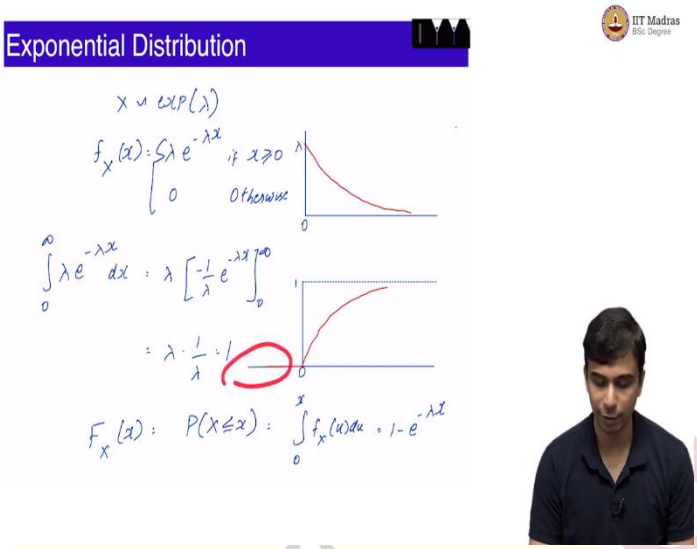
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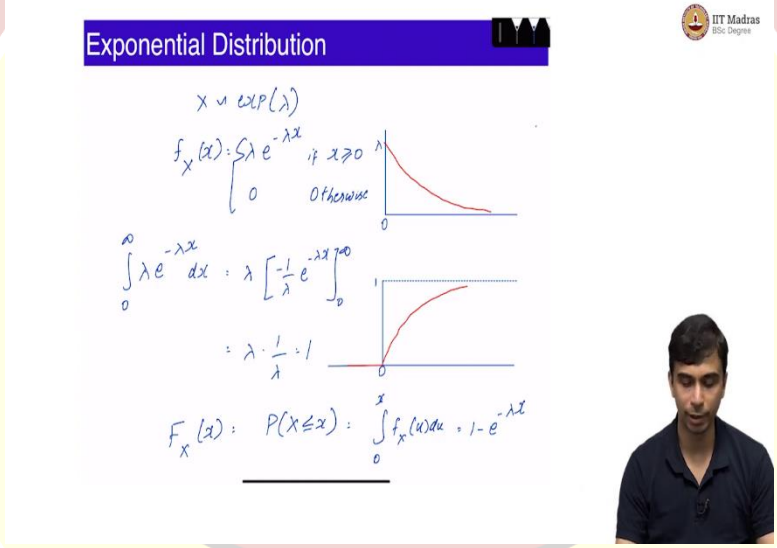
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We will move on to the next distribution, which is called the exponential distribution. The exponential distribution is extremely analogous to the geometric distribution that we saw for the case of discrete random variables. We will first give the notation for that. So, we will say X is distributed as exponential with parameter λ. If that is the case, what is the density function?

Well, $f_X(x) = \lambda e^{-\lambda x}$

f x is equal to if x is greater than equal to 0, and 0 otherwise. So, this is a positive random variable. So, if you plotted the density function of the random variable, this is what it would look like. It could start at 0 and before 0 is going to be 0 $f_X(x)$ is going to decay exponentially. So, $\lambda e^{-\lambda x}$. The first question you should ask for such a definition is that is the valid density or not.

So, for this to be valid density, this must integrate to 1, is that true? Let us do a quick check of that. Let us do a quick check. Integral 0 to infinity $\lambda e^{-\lambda x} dx$ λ - 1 by $\lambda e^{-\lambda x}$ within the range 0 to infinity. Well, this is clearly equal to λ into 1 by λ is equal to 1. So, we know that this, if you integrate the density function from 0 to 1, you get back 1, which means that the area under the curve for this curve is equal to 1.

So, this is a valid density function. But can you get the cumulative distribution function of this random variable as well? You can, and it is a pretty nice function that CDF f X of x is simply probability of X less than or equal to x. Which is indeed exactly equal to integral 0 to x, f X of u du. So, this u is a dummy variable so it does not matter much. And it is straightforward to show that is equal to 1 - e power - λ x.

If x is greater than 0, if x is less than 0, this is going to be equal to f X, the probability of X less than or equal to any negative number is going to be 0, because exponential is a positive random variable. So probability of X less than or equal to X for negative x is going to be 0. And as for positive x is going to be 1 - equals - λ x. So, for that is for x equal to 0, you get me start at 0.

And as x increases, this slowly increases, and it never really hits 1, but it asymptotes to 1. So that is the this 1 here is an asymptote here, it touches the CDF at infinity. Once again, you can see that the left end of this curve is 0 the right end of this curve is infinity. Sorry, right end of this curve is 1, at the left-hand f X of - infinity is 0 and f X of + infinity is equal to 1.

So, this is the basic property of the CDF and this indeed satisfies that. So, what is, why is why am I saying the exponential random variable is similar to the geometric random variable, because this is has very similar properties. And it also is used for modelling a very similar object. You saw that the geometric random variable is very useful for modelling the number of tries before you succeed.

That was the most standard way you would arrive at a geometric random variable. Similarly, the most common uses for exponential random variable is also time, time till failure. For example, a typical example for an exponential random variable might be, you might say that, this the, the running time of this electric bulb is exponentially distributed with λ equal to 0.5. per year.

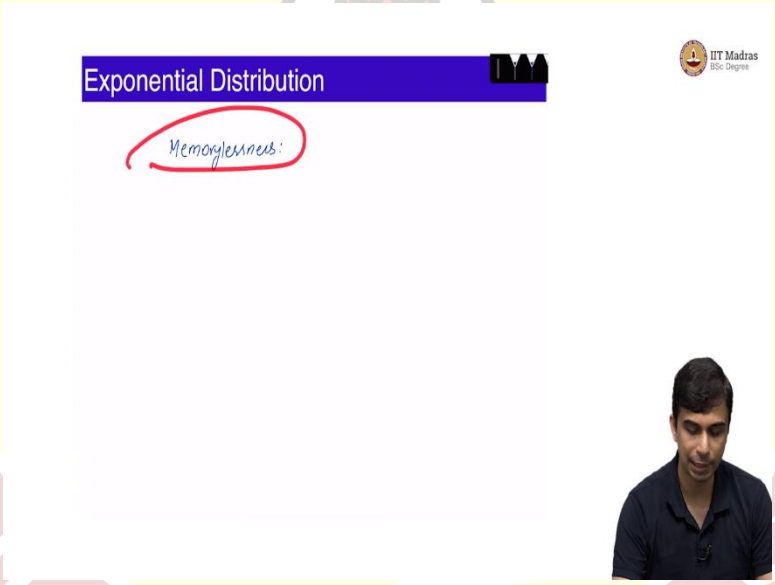
So I am, well, the unit will come to that. So, you might say exponential. What does that mean? That means that you are the they expect the time with which this electric bulb is going to function normally is a random variable, it might fail in the first hour itself, or it might run for

two years, or it might change gradually, the manufacturer needs to know, the random variable needs to know the distribution of the running time of this particular product.

So that based on that, he or she the manufacturer can give a warranty. So, you can say, well, like if I know that for sure this product is going to last for two years, I can safely give out a two-year warranty, and say that, if your product breaks less than two years, I am going to give a replacement. The manufacturer can say that with confidence, because he or she knows that it most likely is going to last two years.

So, this exponential random variable is pretty useful for modelling such kind of objects, which is very similar to the case where exponential random variable is used for modelling time. The same way the geometric random variable is used for modelling the number of tries, the number of tries, you could view it as an analogue of time, where the number of tries is similar to time.

(Refer Slide Time: 08:19)



Exponential Distribution

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Memorylessness: $a > b$

$P(X \geq a | X \geq b) = P(X \geq a-b)$



Exponential Distribution

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Exponential Distribution

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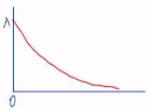
LHS : $\dfrac{P(X \geq a, X \geq b)}{P(X \geq b)}$



Exponential Distribution

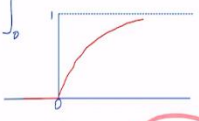
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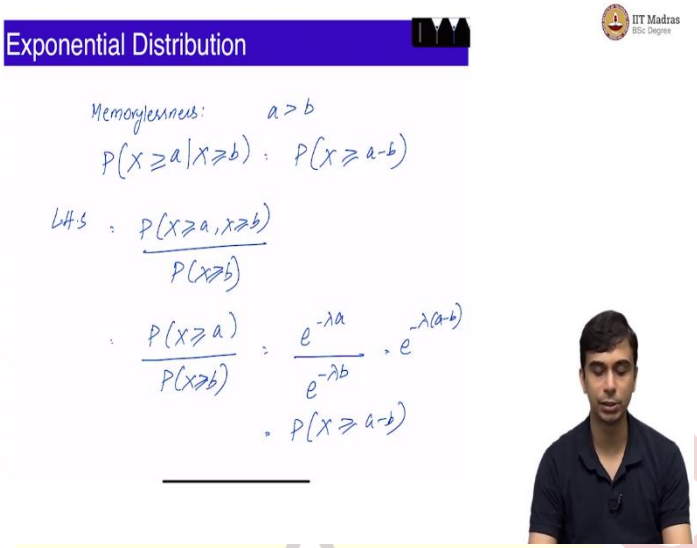
$$\int_0^\infty \lambda e^{-\lambda x}\, dx = \lambda \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_0^\infty$$

$$= \lambda \cdot \frac{1}{\lambda} = 1$$



$$F_X(x) = P(X \leq x) = \int_0^x f_X(u)\, du = 1 - e^{-\lambda x}$$





So, this exponential distribution has certain nice properties. One of the most important properties is what is called us in memorylessness. What does that mean? Let us well, we will just say what the property means. And we will show the satisfies that property. This is what the property says, the probability of x greater than or equal to a given X greater than or equal to b, let us say a is greater than b is equal to probability of X greater than or equal to a - b.

So let us try to interpret this statement. This says that, let us say b is one year and a is three years. What you are saying is, what is the probability that this electric bulb is going to last more than three years, given that it has lasted for one year? That is the situation here, you know that this electric bulb has lasted for one year, you want to know the probability that this electrical bulb is going to last for more than three years.

That is what the memoryless property says that is exactly equal to the probability that this electric bulb would have lasted for more than two years. So, A - B is three - one which is two years. So, this is a natural property that most components in the real world, because each component can fail or succeed with some due to some random event. In that case, exponential satisfies this property and this is a very natural notion.

So, if given the product can be in just two stages. One is its can completely broke down. And second, running perfectly. Like if that is the case, if it is running perfectly for one year, and you want to ask the property, it is going to last for a total of more than three years, it is exactly equal to the probability that from the beginning, it would have lasted for two years. That is what the memoryless property says, let us try to derive this.

What is the left hand side here is probability of X greater than or equal to a and X greater than or equal to b divided by probability of X greater than or equal to b. We have already said a to be greater than b. So probability of X greater than or equal to a and X greater than or equal to b is exactly the same as probability of X greater than or equal to a by probability of X greater than or equal to b.

Probability of X greater than equal to a is exactly equal to 1 - probability of X less than or equal to a, which is $(1 - e^{-\lambda x})$. So, we can say this is equal to e power - λ a and denominator is e power - λ b. Now you can see why you get a - b here; this is equal to $e^{-\lambda} \times (a - b)$. This is exactly equal to the probability that X is greater than or equal to a - b.

So, this is the memoryless property of an exponential random variable. So, you will see a very similar property like this hold also for geometric random variables. But geometry is discrete exponential is continuous. That is the only difference.

(Refer Slide Time: 11:41)



Exponential Distribution

$X \sim exp(\lambda)$

$$E[X] = \int_0^\infty x . \lambda e^{-\lambda x} dx$$



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$$\int u \, dv = uv - \int v \, du$$



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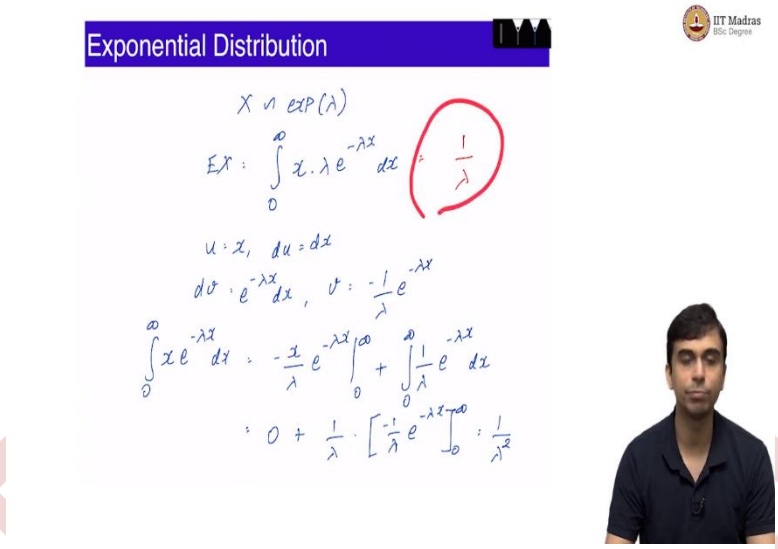
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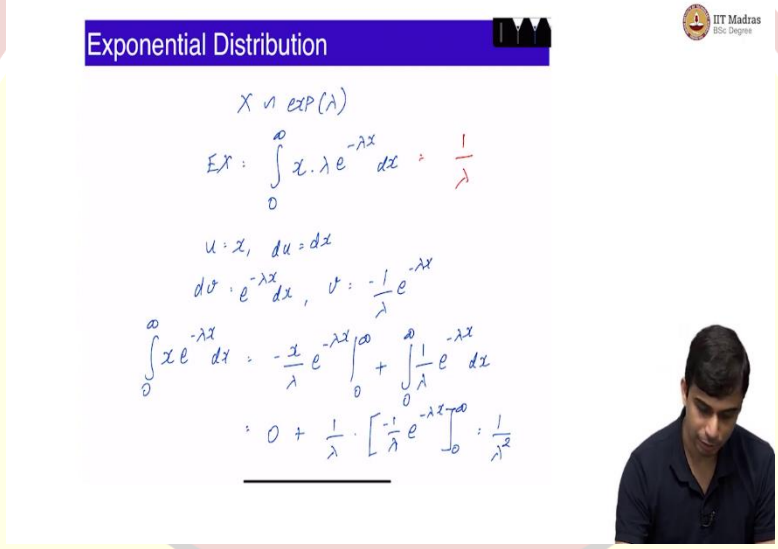
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So, so that is the memoryless property. But before we go to something else, let us first get the expectation and expectation of this particular random variable. What is the expected value of let us say if X is distributed as exponential of λ, what is the expected value of this random variable? Well, this is equal to integral 0 to infinity, x into e power, sorry, x λ e power - λ x dx.

This looks like a complicated integration, but you know exactly what tool to use for these kinds of integrals. That is the integration by parts that you might have seen in your school. Let us go ahead and do that. By integral, what is integration by parts tell you, it says the integral u dv is equal to u v - integral v du. So now, in this particular case, let us set u to be equal to x set u to be equal to x, which means the du is going to be equal to dx and we will set dv to be equal to e power - λ x dx.

So, in which case v would be equal to - 1 /λ e- $^{\lambda x}$ dx. So, now we can write we can do the integration x e power - λ x dx is equal to u v, u is equal to x v is - 1 /λ e- $^{\lambda x}$. So, that would be

- x /λ e- $^{\lambda x}$ integral limits are being 0 to infinity, within limits 0 to infinity - integral v du, v would be - 1 by λ e power - λ x.

So, the - becomes a + here, 1 by λ e power - λ x, du is exactly equal to dx. For v you will not have a dx I am sorry for that. So, v du. Now, let us do this integration. The first term would be 0. Why is that? Because substitute infinity looks like infinity into something but e power - λ x is much, it is much closer to 0 than x. And if x keeps on increasing, x e power - λ x goes to 0 because e power - λ x is much closer to 0 than x is to infinity.

So, x e power - λ x is the limit as x goes to infinity is 0, and 0 e power - λ 0 is also 0. So, this term is entirely equal to 0. What about the term on the second term here? That is +. So that is 1 by λ equal - λ x. 0 + 1 by λ into e power - λ x, which is going to be equal to - 1 by λ e power - λ x within the range of 0 to infinity.

So, you get, this, you already have 1 by λ here, and - 1 by λ - λ x in the range of 0 to infinity, which is equal to 1 by λ e power - λ x in the range of infinity to 0, which would be equal to 1 by λ square. So, what we have is integral 0 to infinity, x e power - λ x is equal to 1 by λ square. But what did we want, we want this integration, which is simply λ times this integration.

So, 1 by λ square into λ, which is equal to 1 by λ. So, that means that x is equal to 1, you can see the similarities to the geometric random variable coming here again. So, you saw that the for the expectation of a geometric random variable was 1 by P, here, 1 by λ. So that is similar here, if you have a very small λ, which means, then you have EX to be quite large.

So, λ unit of λ for example, in the case of bulb failure experiment would be per year. So, if you have λ of let us say 0.5 per year, then you expect this bulb to last for two years. So, that is the exponential expectation of the exponential random variable. You can define, you can derive the variance also using the same way using integration by parts. But the variance is not very important right now. So, the expected value of an exponential random variable is nicely equal to 1 by λ.

(Refer Slide Time: 16:38)

ponential Distribution

$X \sim exp(\lambda)$

$Y \sim exp(\tau)$

$Z = min(X, Y)$

S.T: $Z \sim exp(\lambda+\tau)$



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S.T: $Z \sim exp(\lambda+\tau)$



ponential Distribution

$X \sim exp(\lambda)$ $0.25\ year^{-1}$

$Y \sim exp(\tau)$ $0.25\ year^{-1}$

$Z = min(X, Y)$

S.T: $Z \sim exp(\lambda+\tau)$ $0.5\ year^{-1}$



ponential Distribution

$X \sim exp(\lambda)$ $0.25\ year^{-1}$

$Y \sim exp(\tau)$ $0.25\ year^{-1}$

$Z = min(X, Y)$

S.T: $Z \sim exp(\lambda+\tau)$ $0.5\ year^{-1}$

$F_Z(z) = 1 - (1 - F_X(z))(1 - F_Y(z))$

$= 1 - ($



ponential Distribution

$X \sim exp(\lambda)$ $\quad 0.25\ year^{-1}$

$Y \sim exp(\tau)$ $\quad 0.25\ year^{-1}$

$Z = min(X, Y)$

S.T: $\quad Z \sim exp(\lambda+\tau)$ $\quad 0.5\ year^{-1}$

$$F_Z(y) = 1 - (1-F_X(y))(1-F_Y(y))$$
$$= 1 - (e^{-\lambda y})(e^{-\tau y}) \quad \text{if } y \geq$$
$$= 1 - e^{-(\lambda+\tau)y}$$



ponential Distribution

$X \sim exp(\lambda)$ $\quad 0.25\ year^{-1}$

$Y \sim exp(\tau)$ $\quad 0.25\ year^{-1}$

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$$= 1 - e^{-(\lambda+\tau)y}$$

$$f_Z(y) = \begin{cases} 0 & \text{if } y < 0 \\ (\lambda+\tau)e^{-(\lambda+\tau)y} & 0 \cdot w \end{cases}$$



ponential Distribution

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Let us go ahead and X is distributed as exponential with parameter λ. And let us say Y is distributed as exponential with parameter, let us say tau and x and we say we will say X and Y are independent. Now we define Z to be equal to minimum of X , Y. Then what is the distribution? What is the what is, what we will show is that z is distributed, we will show that Z is distributed as exponential with parameter given by λ + tau.

This is particularly nice, because you can interpret this X and Y. Let us say you have a CPU, or you have a laptop, and there are only two components, let us see CPU and RAM. See, the failure time of the CPU is exponential with the 1 by 4 per year, 0.25 per year. And let us say Y the RAM is a failure. The failure time of the RAM is exponential with, let us say, another 1 by 4 per year.

So effectively, you individually you expect your RAM to last four years and CPU to last four years. But you give a warranty to the laptop as a whole, then how long should you give the warranty for? Let us say, what you do is you just replace the warranty, we just replace the laptop as a whole. If that is the case, we have two components. And what matters is even if one of them fails, you have to, let us say, junk the entire thing.

If that is the case, z would essentially correspond to the failure time of the laptop as a whole. The meaning of if, whenever the first the first, whenever the first of these two things fail, the entire thing fails. And so Z will mean that X , Y. What the this particular simple theorem states that Z the failure rate of Z, Z is also an exponential random variable with λ + tau. And that is natural since if we have X to be the failure rate of let us say 0.25 per year and Y to be 0.25 per year also, then z would be 0.5 per year.

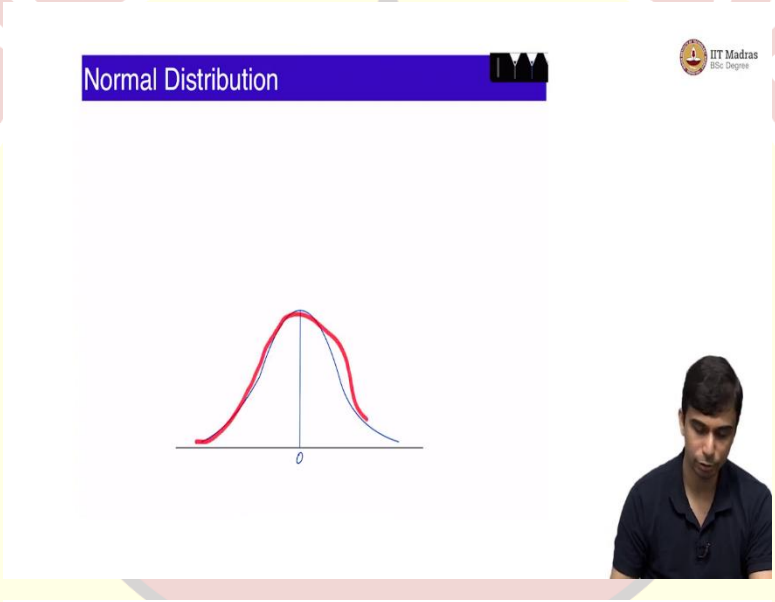
So, which means that you the expected value of Z is going to be equal to 2. That is not particularly, that is intuitively obvious, but we have to prove this too. How do we prove that Z is distributed as exponential of λ + tau when Z is equal to mean of X , Y, where X and Y are independent?

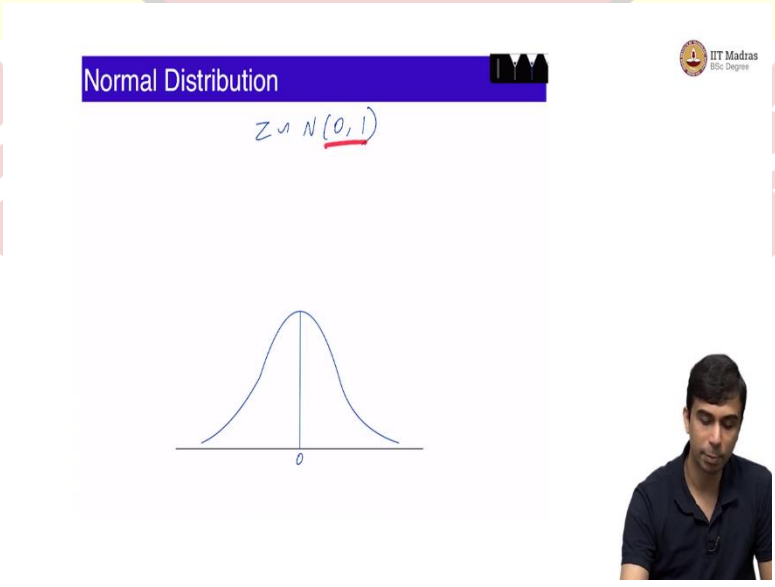
We already have the formula for the minimum of how do we get the density of the minimum of two independent random variables and based on that we have F this CDF of Z, F z of Z is simply equal to 1 - 1 - f X of Z into 1 - f Y of Z. This we derived previously for general minimum of two random variables, you can substitute that here. This is 1 -; f X of Z is already 1 - e power - 1 And 1 - 1 - that will simply be e power - λ Z.

And this would be e power - tau Z. This is equal to 1 - e power - λ + tau Z. So, if Z is greater than equal to 0. This thing is, if Z is greater than equal to 0. If Z is less than 0, then it is clearly 0. Because your CDF of Z cannot mean this Z cannot take negative value. So, F z of Z, if z is less than 0 is clearly 0. Now what is 1 - e power - λ + tau Z.

That is exactly equal to the which the which means that the density of Z, f Z of Z is equal to 0. If Z less than 0, and λ + tau into e power - λ + tau Z otherwise. So, this is exactly the density of an exponential random variable with parameter λ + tau. So, this is one way we can see how exponential random variable is a very natural way for modelling time have failure.

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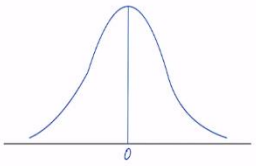




Normal Distribution

$Z \sim N(0,1)$

$$f_Z(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right)$$

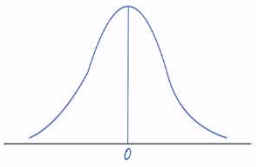




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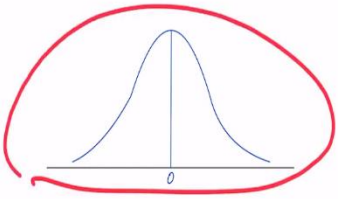




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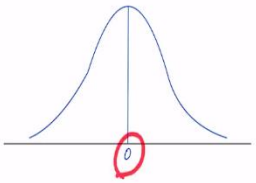




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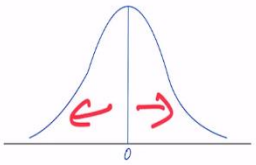




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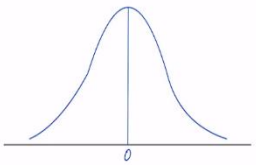




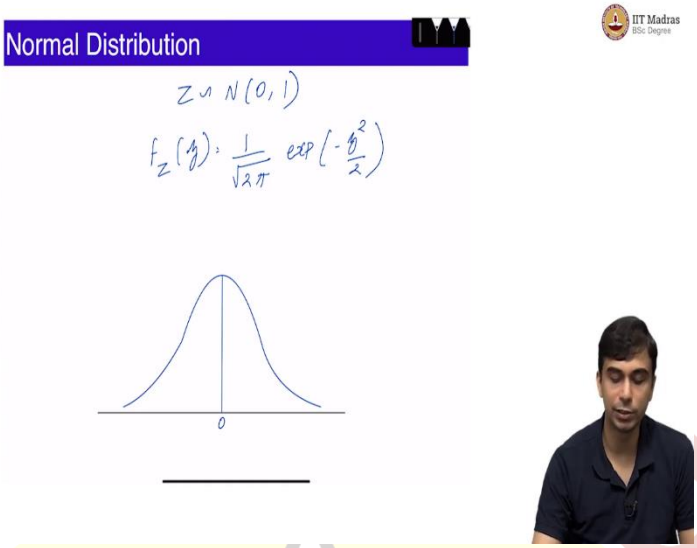
Normal Distribution

$Z \sim N(0,1)$

$f_Z(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right)$





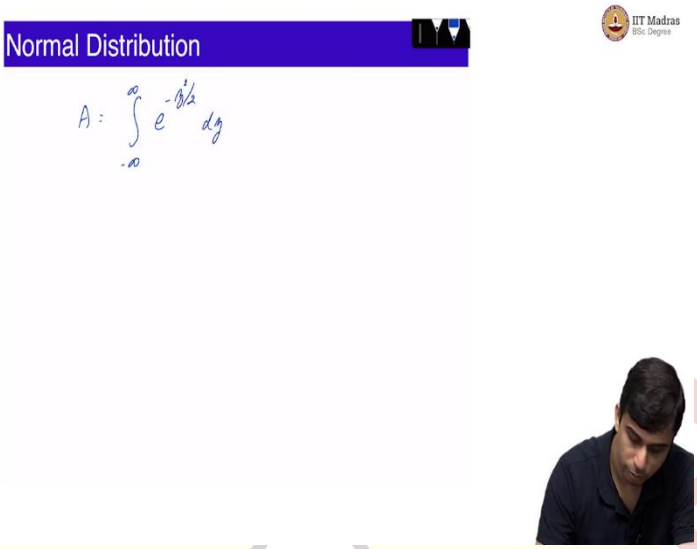


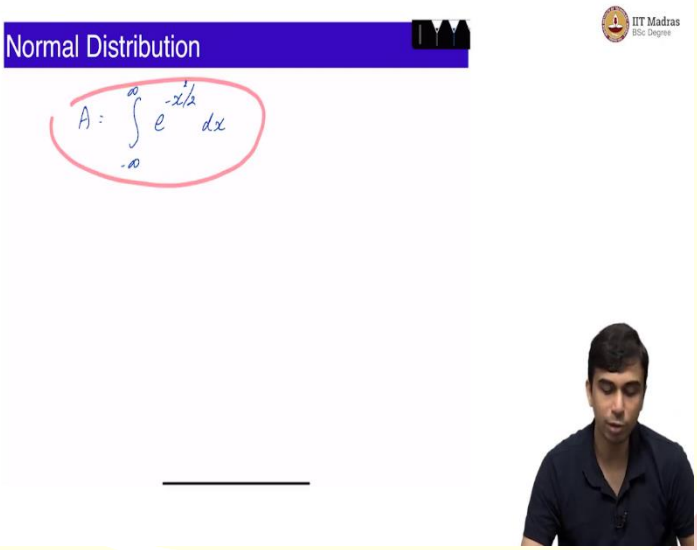
So now we will move to the next example, which is the normal distribution. The normal distribution is something which you might have seen already in your school. And it has a very characteristic bell curve. So, the density of the normal distribution, so we will say Z is distributed as normal standard normal or normal with parameters 0 and 1, we will say what the general parameters are pretty soon.

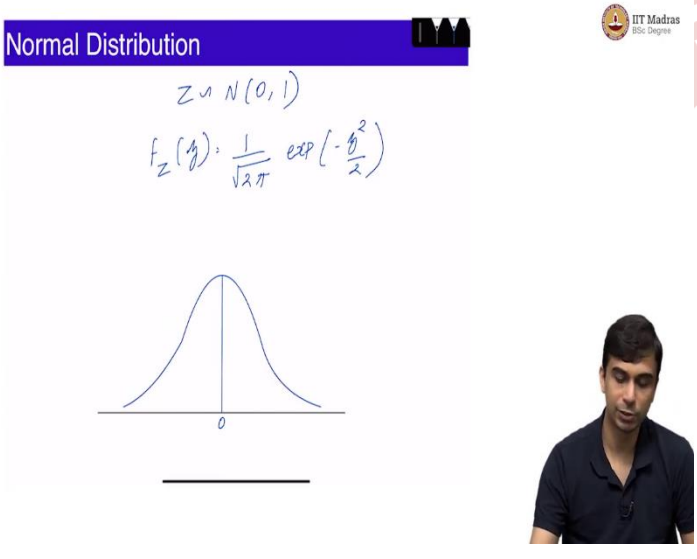
But in the case of the standard normal, which corresponds to saying that z is distributed as normal with parameters 0 and 1, we will say that f the density Z, F Z of Z is equal to 1 by square root of 2 pi x of - Z square by 2. This is the density function of the standard normal random variable. And if you plotted the density function, you would have this characteristic bell-shaped curve.

It is implicit centred around 0. And it decays in this way, as you move either in the negative direction, or in the positive direction. So Z can take both positive or negative values. So, this is the density function of the standard normal. So, first question you should immediately ask is, Is this a valid density? How do you know there is a valid density? Well, you have to integrate, and this from 0 to sorry, - infinity to infinity, and check if it is valid density.

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Normal Distribution

$$A = \int_{-\infty}^{\infty} e^{-x^2/2}\, dx$$



Normal Distribution

$$A = \int_{-\infty}^{\infty} e^{-x^2/2}\, dx$$

$$A^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2}\, dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}\, dy\right)$$



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Normal Distribution

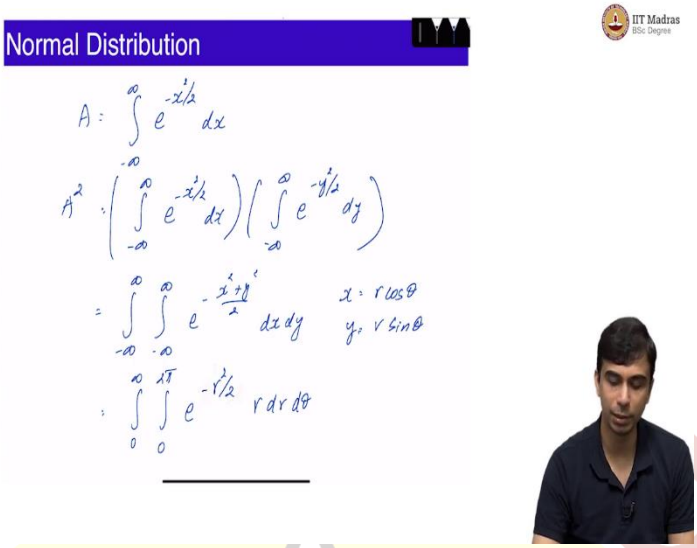
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$$= \int_{0}^{\infty}\int_{0}^{2\pi} e^{-r^2/2}\,r\,dr\,d\theta$$





So now for that, let us go ahead and do this. Let us define some integral here, A to be equal to integral - infinity to infinity, e power - z square by 2 dz. Let us use x here, it would be easy for us to use x here. How do you? Essentially, we want this for if our density here was a valid density? If you integrated this from - infinity to infinity, this should you should have you should get root 2 pi.

How would you get square root of 2 pi? Well, let us try and do this integration. So, in fact, this integration is quite hard to do. But there is a small trick that we can do to make this much easier integration. Let us define, let us instead of doing for A, let us do A square and say a square is equal to integral - infinity to infinity e power - x square by 2 dx into integral - infinity to infinity e power - y square by 2 dy.

Why am I doing this? y is a dummy variable so it does not matter here. This is also equal e is also equal to A, but we can bring both these together into a single integral we will call integral - infinity to infinity e power - x square + y square by 2 dx dy. How does this actually help us? The reason this helps us is because we will now can do a change of variable here.

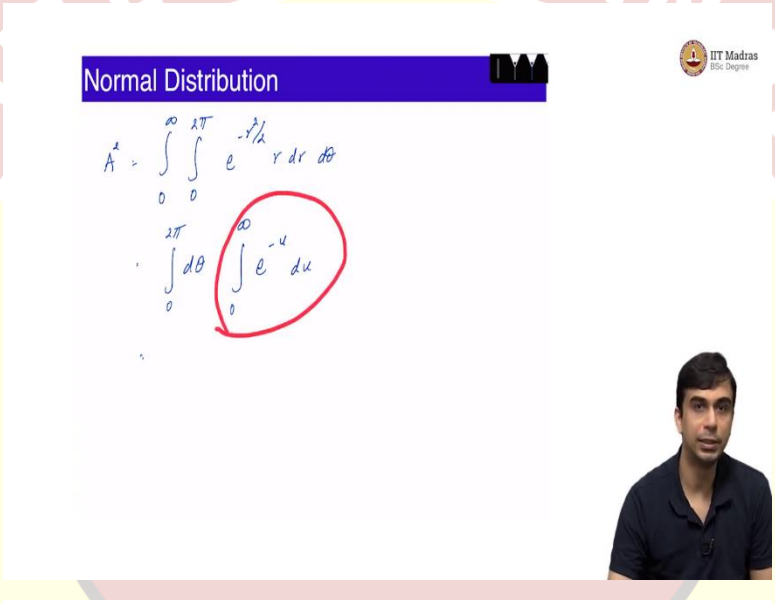
You can view this as integral integrating e power - x square + y square by 2 over the entire XY plane. So now we can use a cartesian to polar transformation where we will define r and theta as follows where x is equal to r cos theta, and y is equal to r sin theta, the standard polar coordinate transformation. So, integrating, over x and the entire XY plane exactly corresponds to integrating over r from 0 to infinity and theta from 0 to 2 pi.

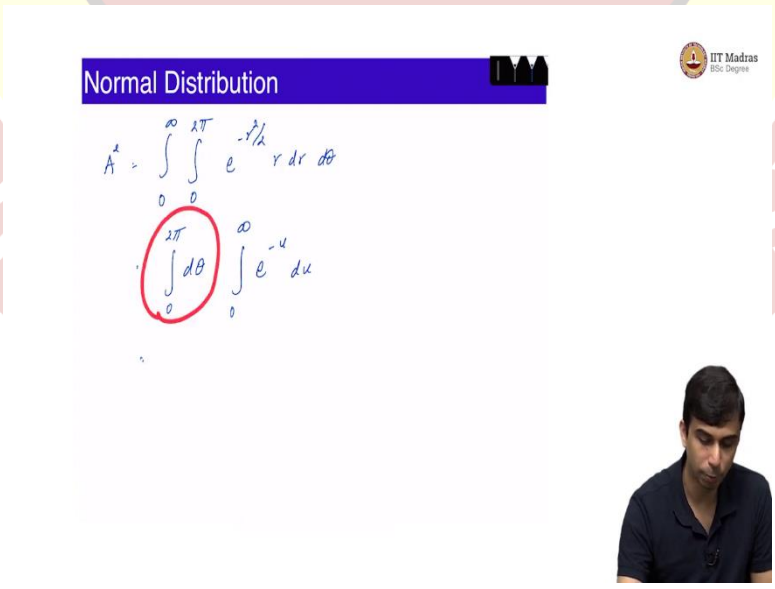
Of course, dx dy has to do change of variable rule dx dy we know happens corresponds to r dr d theta. This is also standard integration trick, integration rate of change of variable that you

might have seen. So, you vary are from the range 0 to infinity and theta from the range 0 to 2 pi. But why do we do that? Well, that helps us because x square + y square is exactly equal to r square.

This becomes e power - r square by 2 and dx dy simply becomes r dr d theta. The main advantage of this particular change of variable is that now you have r square and r dr. So now we can do another change of variable where r square by 2 becomes the variable and r dr. We can get it out. Let us do that integration.

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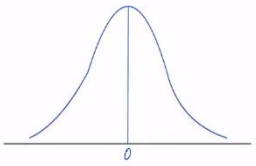




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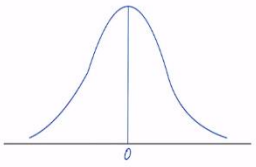




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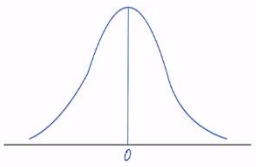




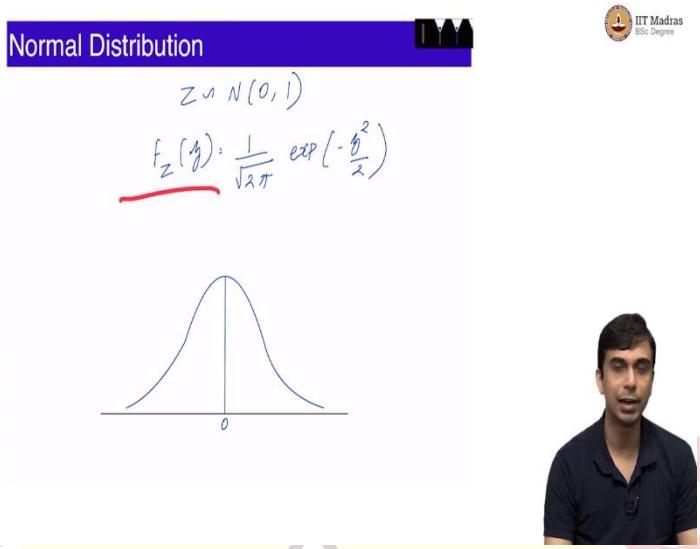
Normal Distribution

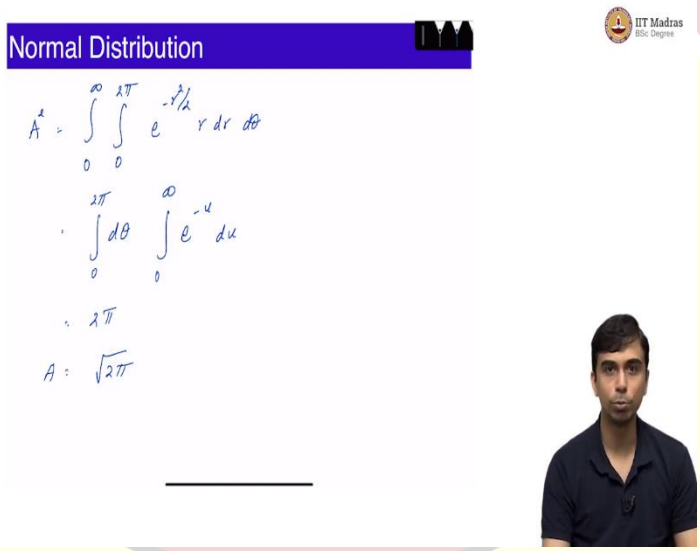
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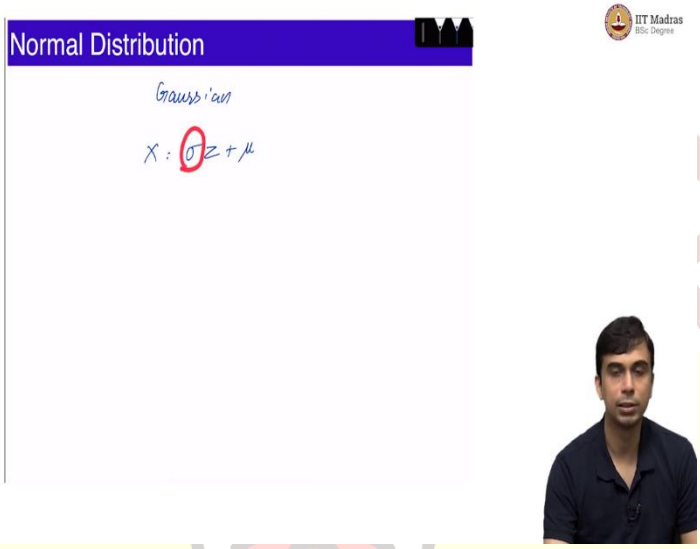


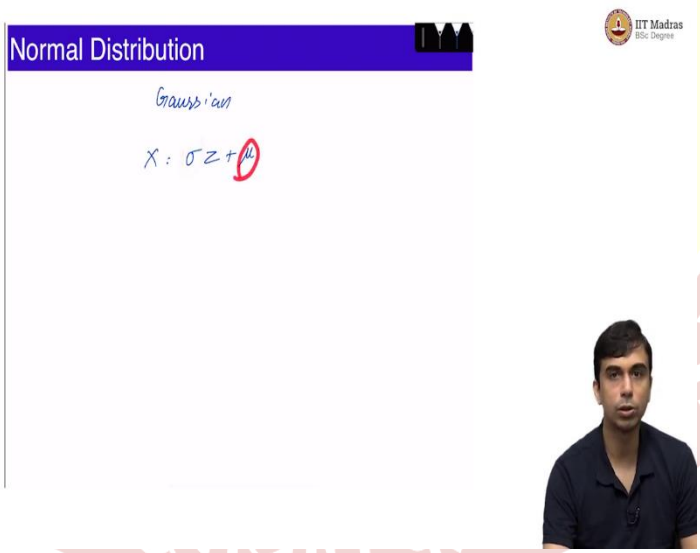
Integral 0 to A square is equal to integral 0 to infinity, integral 0 to 2 pi e power - r square by 2 r dr d theta. So, you can take the d theta integration outside. So, it is integral 0 to 2 pi d theta integral 0 to infinity, where instead of r square by 2, let us define u to be equal to r square r square by 2. So, if you do that, the limit of u does not changes to 0 to infinity. But du is exactly equal to r dr.

So, you have e power - u, du. And this integration is pretty easy, you can do that, and you will get back one. And this integration, this integration is also easy. It is equal to 2 pi. So, we get A square is equal to 2 pi, which means that A is equal to square root of 2 pi. Which means that our density is a valid density, because if you integrate this, from - infinity to infinity, you get square root 2 pi.

So that is why you have this 1 by square root 2 pi to cancel it out. So now we have that if you integrate this from - infinity to infinity, you get the value of 1, which means that this is a valid density. So, this settles that this is a valid density. Why are we actually bothering so much about this particular distribution?

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Normal Distribution

Gaussian

$X = \sigma Z + \mu$



Normal Distribution

Gaussian $\quad Z \sim N(0,1)$

$X = \sigma Z + \mu$

$f_X(x) = ?$



Normal Distribution

Gaussian $\quad Z \sim N(0,1)$

$X = \sigma Z + \mu \quad , \quad Z = \frac{X-\mu}{\sigma}$

$f_X(x) = ?$

$= f_Z\left(\frac{x-\mu}{\sigma}\right).$



Normal Distribution

Gaussian $Z \sim N(0,1)$

$X = \sigma Z + \mu$; $Z = \frac{X-\mu}{\sigma}$

$f_X(x) = ?$

$= f_Z\left(\frac{x-\mu}{\sigma}\right)$.



Normal Distribution

Gaussian $Z \sim N(0,1)$

$X = \sigma Z + \mu$; $Z = \frac{X-\mu}{\sigma}$

$f_X(x) = ?$

$= f_Z\left(\frac{x-\mu}{\sigma}\right) \cdot \frac{1}{\sigma}$



Normal Distribution

Gaussian $Z \sim N(0,1)$

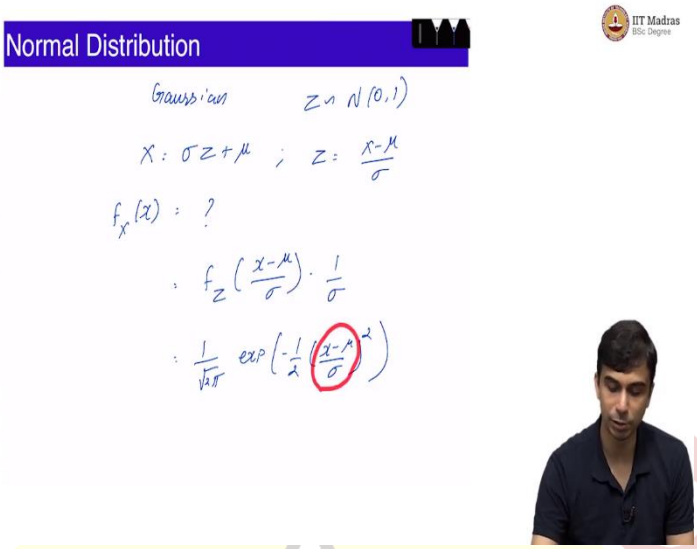
$X = \sigma Z + \mu$; $Z = \frac{X-\mu}{\sigma}$

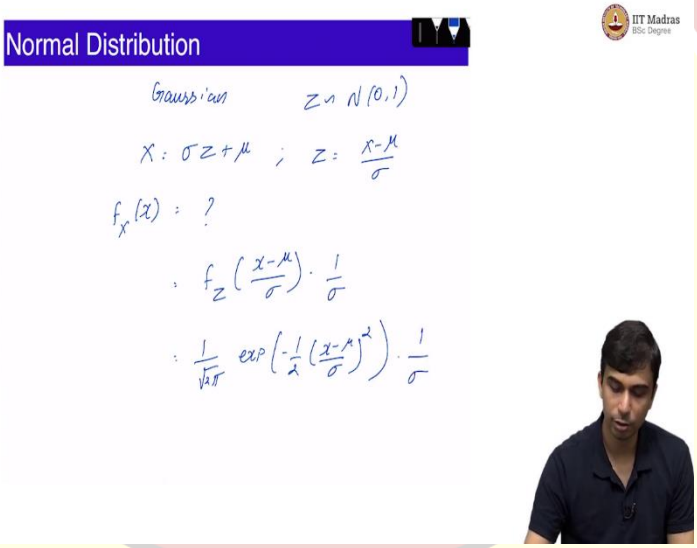
$f_X(x) = ?$

$= f_Z\left(\frac{x-\mu}{\sigma}\right) \cdot \frac{1}{\sigma}$

$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$







Because you will soon see that this is a very commonly occurring distribution in nature. And one of the main reasons is what is called a central limit theorem, which we will soon see that practically, all situations where multiple effects are essentially added together, and you see an effect, you can expect the effect to be a normal distribution. There is another very name for a normal distribution just called the Gaussian distribution.

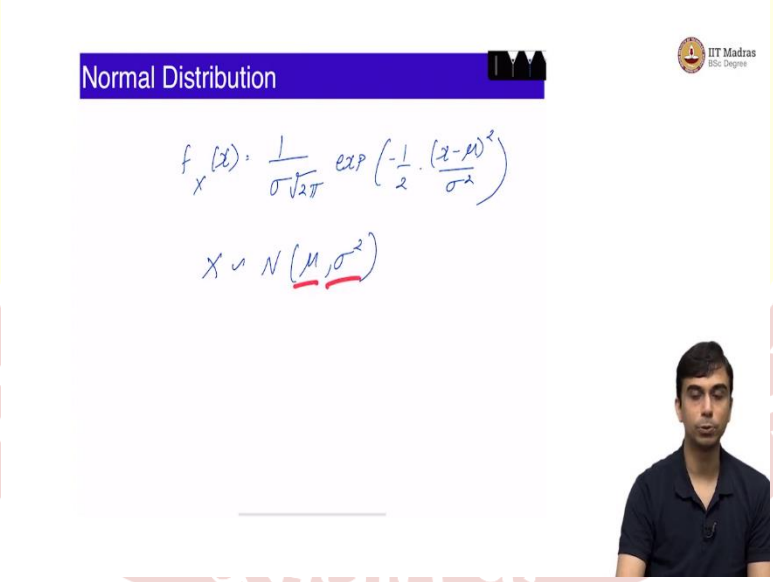
It is also called the Gaussian. So, we will see the central limit theorem soon. But there are several practical examples of normal distribution. For example, let us say you are measuring current in a wire when there is no voltage difference. Let us say you have a very accurate ammeter. And you are measuring the current in a wire, what you are seeing is just noise, and that noise is normally distributed, or Gaussian distributed, and so on.

You will see several examples in practice where things are normally distributed. So, we saw Z distributed as normal 0, 1. But what are these two parameters 0 and 1. We will see that soon. Now, let us define x to be equal to sigma times Z + mu. So, sigma is a scalar constant and mu is also a scalar constant. So, x is equal to sigma Z + mu. Now what can we let us say z is distributed as normal 0, 1.

What can we say about the density of x? So, f X of x is equal to what? That is the question. You can use or change a variable rule, because this happens to be an invertible transform. So, f X of if x equal to sigma Z + mu z equal to X - mu by sigma. So, f X of x is simply equal to f Z of x - mu by sigma and then you have the Jacobean of this transformation. So, it is just a 1 to 1 thing, but you can still do the Jacobian of this, which happens to be just equal to 1 by sigma.

So, what is f Z of x - mu by sigma, we can just substitute that here. This is 1 by square root of 2 pi exp of - half x - mu by sigma the whole square. So, I have just replaced z by x - mu by sigma. So, this is x - mu by sigma, the whole square and then you have a sigma.

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Normal Distribution

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\cdot\frac{(x-\mu)^2}{\sigma^2}\right)$$

$X \sim N(\mu, \sigma^2)$

$EX = ?$



Normal Distribution

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\cdot\frac{(x-\mu)^2}{\sigma^2}\right)$$

$X \sim N(\mu, \sigma^2)$ $\quad X = \sigma Z + \mu$

$EX = ? = E[\sigma Z + \mu]$

$= \mu + \sigma E[Z]$



Normal Distribution

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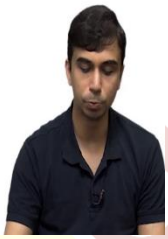


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$$= \mu$$
$$Var[X] = ?$$



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$$EX = ? = E[\sigma Z + \mu]$$
$$= \mu + \sigma E[Z]$$
$$= \mu$$
$$Var[X] = ? = \sigma^2 Var[Z]$$



Normal Distribution

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$$Var[X] = ? \quad , \quad \sigma^2 \cdot Var[Z] = \sigma^2$$



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$= \mu + \sigma E[Z]$

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$Var[X] = ? \; , \; \sigma^2 . Var[Z] = \sigma^2$



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$EX = ? = E[\sigma Z + \mu]$

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$Var[X] = ? \; , \; \sigma^2 . Var[Z] = \sigma^2$



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$$= \mu$$

$$Var[X] = ? \; , \; \sigma^2 \cdot Var[Z] = \sigma^2$$



So, put these together and you get a standard definition of a normal distribution that is f X of x is equal to 1 by sigma square root of 2 pi exp of - 1 by 2x - mu the whole square by sigma square. And we will just say X is distributed as normal with mu and sigma square. So, now we get the general normal random variable instead of just 0 and 1. But mu and sigma square we get back.

So, what is the expected value of EX? You could in principle just integrate this over - infinity to infinity and get the answer, but you do not need to. Why? Because X itself was distributed as the sigma z + mu. So, expected value of x is simply equal to expected value of sigma z + mu. And by linearity of expectation, we have Z equal to mu + sigma into expected value of Z.

What is the expected value of Z? This is pretty easy, because z is actually a symmetric random variable around 0. So, expected value expected value of X is equal to 0. So, this is equal to mu. So, if you have a normal mu, sigma square random distribution, the mean is new. What about the variance? What about the variance of x? Well, you can use the same sigma Z + mu effect here to get this is equal to sigma square into variance of Z.

What is variance of Z? What is variance of Z. I have said you have to do an integration, which is you have to integrate that Z X square e power - x square by 2 within the range of - infinity, Infinity. But what you will get is where in subset is actually equal to 1 that is a single integration, I am not going to do that, but that is equal to 1. So, this is equal to sigma square. So, variance of x is equal to sigma square and expected value of x is equal to mu.

So that is why you will often see these mu and sigma square be referred to as the mean parameter and the variance parameter. So, mu is exactly equal to the mean and sigma square is exactly equal to the variance. The mean and expectation will be used interchangeably as I mentioned earlier. So, this is called mu is the mean parameter and sigma square is the variance parameter. So, with that, we are done with the standard distributions that we will see which form the building blocks for more complex distributions.